

sim-to-real boundary

TRAINING - SUMO

Prior RL-TSC

privileged state

exact queues, per-veh waiting, net throughput

privileged reward

delay, arrived vehicles, global throughput

Ours

camera-computable state

26-dim: queue, occupancy, speed, class shares, pressure

reward from the same proxies

queue • pressure • throughput
switch • low-speed • waiting

DEPLOYMENT - CAMERA

camera cannot measure
(observation gap)

camera cannot provide
(reward gap)

YOLOv11 + ByteTrack

→ same 26-dim obs

→ actor π_θ → phase

reward also computable

on-camera — zero
privileged exception

identical schema

same proxy terms